

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	30 September 2026
Team ID	SWTID-2026-1429
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Story Input & Customization	Accept story prompt, main character name, setting, story tone and art style through the web form.
FR-2	AI Story Generation	Generate a structured 5-panel outline with Gemini Flash, then generate detailed narration and character dialogue with Gemini Pro.
FR-3	AI Illustration & Preview	Generate a comic-style illustration for each panel using Stable Diffusion; match images with panel text and display the comic sequentially.
FR-4	Comic Layout & PDF Export	Build the final panel layout and compile images, titles, descriptions, captions and narration into a downloadable multi-page PDF using FPDF.
FR-5	API Access & Testing	Provide FastAPI routes for browser form generation, JSON-based generation and direct image-generation testing.
FR-6	Future Extensibility	Maintain modular components so future enhancements such as user profiles, comic libraries and multi-page story arcs can be added.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple web interface with clear inputs and sequential comic preview; designed to be accessible to non-artists and casual users.
NFR-2	Security	Sensitive Gemini and Hugging Face credentials are stored as environment variables rather than hard-coded in application code.
NFR-3	Reliability	Modular generation pipeline with backend error handling using HTTPException and try-except blocks for unexpected failures.
NFR-4	Performance	Use Gemini Flash for fast structured outlines and Gemini Pro for richer narrative generation; generate only the required panel images.
NFR-5	Availability	Local FastAPI deployment is supported; the architecture is designed for future cloud deployment.

NFR-6	Scalability	Separated frontend, FastAPI orchestration, AI modules, layout builder and exporter allow components to be extended independently.
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